

SC2014 : Sensors, Interfacing & Digital Control

Part 2 : Tutorial 4 – Transient Analysis & Stability

Q1 : The transfer function of an open-loop chemical plant, $G(z)$, is given here. A controller function, $C(z)$, is also given. Design a closed-loop unity gain negative feedback system for this plant and the controller. Find the Controller Gain, K , for the system to have a damping ratio of 0.3, a natural undamped frequency of 70 rad/s and a sampling rate of 20 samples/sec.

Transfer function = $GC/(1+GCH)$, H is the sensor

Get the 2nd order characteristic equation (no need remember)

$$G(z) = \frac{1}{z^2 + 0.686z + 0.002} \quad C(z) = \frac{K}{200}$$

sub in damping = 0.3, $\omega_n = 70$, $T = 1/20 = 0.05$, end up the characteristic eqn = $z^2 + az + b$, see phone gallery (6 Nov)

Q2 : The system characteristic equation of the mechanical system is given as :

Characteristic eqn: $P(z) = z^2 - 1.321z + 0.497$

Find the damping ratio, ζ , and undamped natural frequency, ω_n for this system when the sampling period is 0.2s.

SEE PHONE GALLERY 6 Nov.

For Question 1 to 2. You may assume the standard characteristic equation for a second order system is :

$$z^2 - 2 \cos(\omega_n T \sqrt{1 - \delta^2}) e^{-\delta \omega_n T} z + e^{-2\delta \omega_n T}$$

Q3 : Examine the stability of the characteristic equation given by :

$$P(z) = (z - 1)(z - 0.8)(z + 0.5)$$

Plot the poles on the z -plane and comments on the positions of the poles and how they affect the stability of a system.

Apply Jury's stability test and comment on the stability of the system.

SEE PHONE, Qns no. + 1, Q3 here is Q4 on phone etc.

Q4 : Consider the system described by :

$$y(k) + 2.1y(k - 1) + 0.4y(k - 2) = x(k) + x(k - 1)$$

where $y(k)$, $x(k)$ are output/input of the system. Determine the stability using Jury's stability test.